

Capture the Flag 2: Widgets

Goal: Turn the single-card application into a simple multi-section interface.

Note from the instructor before you start:

These activities involve using your own problem solving and researching skills. We are strictly to implement NOT TO USE AI to generate the code for you. You may only use AI for RESEARCHING the syntax or usage, but NEVER THE CODE ITSELF.

Consequence: Immediate 0 score in your grade for this activity.

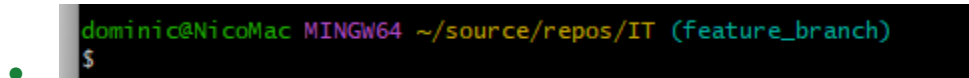
(I know what AI-generated code looks like, don't even think about it :>)

Flag 0 – Create Your Feature Branch

Objective: Create a feature branch before modifying your application. Name the branch feature/widgets

Screenshot:

- Terminal showing `feature/widgets`



Flag 1 – Break It Apart

Objective: Separate your profile information into individual `Text` widgets.

If your card currently looks like this:

`Text(`

`'Chiara Marie C. Canque\n'`

'BSIT - 3\n'

'My First Flutter Application\n'

'August 11, 2026',

)

Your task: Separate each piece of information into its own `Text` widget.

You should have something conceptually like:

Text → Name

Text → Course & Section

Text → Application Title

Text → Date

Do not worry about making it pretty yet.

Screenshot:

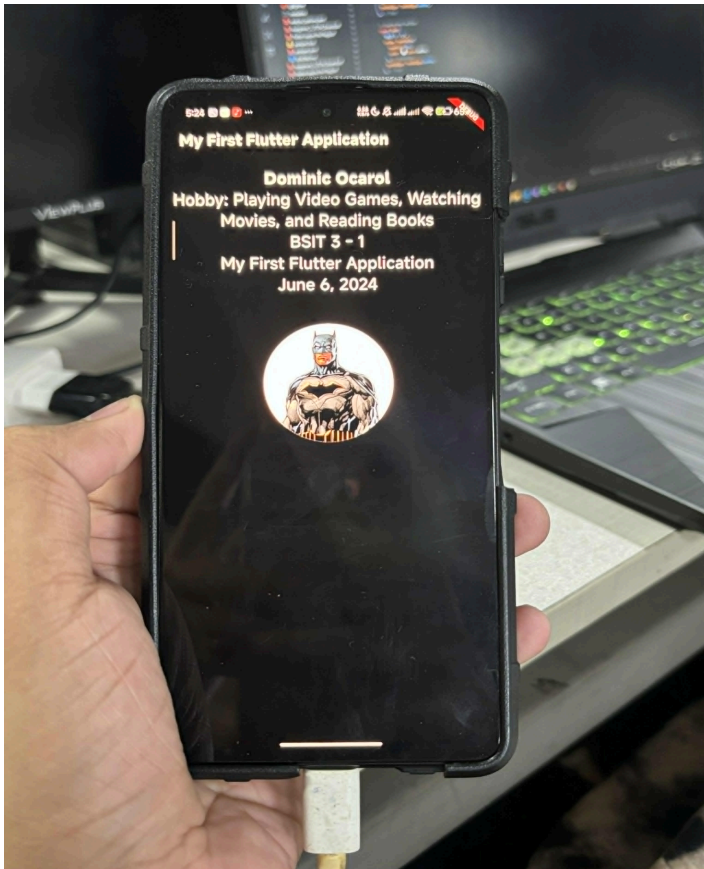
- Your code showing the separate `Text` widgets
- Your application

```
//Name
const Text(
  'Dominic Ocarol',
  style: TextStyle(
    fontSize: 24,
    fontWeight: FontWeight.bold,
    color: Colors.white,
  ), // TextStyle
  textAlign: TextAlign.center,
), // Text

//Course and section
const Text(
  'BSIT 3 - 1',
  style: TextStyle(
    fontSize: 18,
    color: Colors.white,
  ), // TextStyle
  textAlign: TextAlign.center,
), // Text

//Application title
const Text(
  'My First Flutter Application',
  style: TextStyle(
    fontSize: 18,
    color: Colors.white,
  ), // TextStyle
  textAlign: TextAlign.center,
), // Text

//Date
const Text(
  'June 6, 2024',
  style: TextStyle(
    fontSize: 18,
    color: Colors.white,
  ), // TextStyle
  textAlign: TextAlign.center,
), // Text
```



🚩 Flag 2 – Stack the Information

Objective: Arrange your individual `Text` widgets vertically.

Research how to use:

`Column()`

Place your information inside the `Column`.

Your card should now have:

Name

Course & Section

Application Title

Date

Screenshot:

- The `Column` in your code
- The four separate pieces of information displayed vertically

```
child: Column(  
  mainAxisAlignment: MainAxisAlignment.start,  
  children: [  
    const SizedBox(height: 20),  
  
    //Name  
    const Text(  
      'Dominic Ocarol',  
      style: TextStyle(  
        fontSize: 24,  
        fontWeight: FontWeight.bold,  
        color: Colors.white,  
      ), // TextStyle  
      textAlign: TextAlign.center,  
    ), // Text  
  
    //Course and section  
    const Text(  
      'BSIT 3 - 1',  
      style: TextStyle(  
        fontSize: 18,  
        color: Colors.white,  
      ), // TextStyle  
      textAlign: TextAlign.center,  
    ), // Text  
  
    //Application title  
    const Text(  
      'My First Flutter Application',  
      style: TextStyle(  
        fontSize: 18,  
        color: Colors.white,  
      ), // TextStyle  
      textAlign: TextAlign.center,  
    ), // Text  
  
    //Date  
    const Text(  
      'June 6, 2024',  
      style: TextStyle(  
        fontSize: 18,  
        color: Colors.white,  
      ), // TextStyle  
      textAlign: TextAlign.center,  
    ), // Text  
  ],  
)
```


🚩 Flag 3 – Add Another Element

Objective: Add another individual widget to your `Column`.

Add something that belongs to your profile.

Examples:

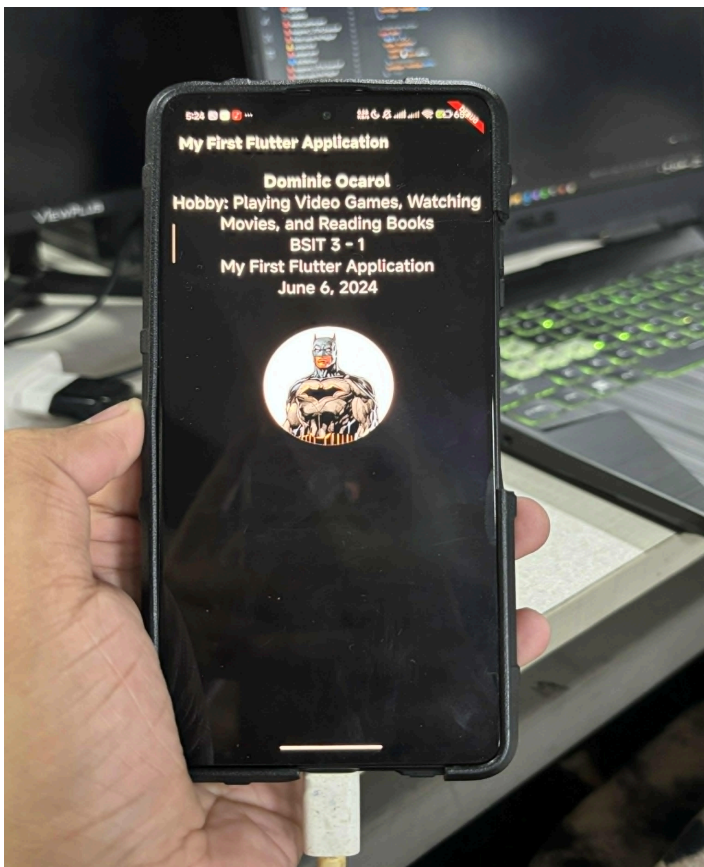
Hobby

It must be its **own widget**, not additional text inside another `Text`.

Screenshot:

- New widget in your `Column`
- Updated application

```
5
6 // Hobby
7 const Text(
8   'Hobby: Playing Video Games, Watching Movies, and Reading Books',
9   style: TextStyle(
10    fontSize: 24,
11    color: Colors.white,
12  ), // TextStyle
13  textAlign: TextAlign.center,
14), // Text
```



Flag 4 – Put Two Things Side-by-Side

Objective: Discover the `Row` widget.

Create a new section containing **two pieces of information beside each other**.

For example:

Age: 21 Birthdate: August 11, 2003

Research:

`Row()`

Place the `Row` inside your existing `Column`.

Your structure should now resemble:

Column

├── Text

├── Text

├── Text

├── Text

├── Text

└── Row

├── Text

└── Text

Screenshot:

- Your widget structure/code
- The two items displayed side-by-side
-

```
Row(  
  mainAxisAlignment: MainAxisAlignment.center,  
  children: [  
    const Text(  
      'Dominic Ocarol\n',  
      style: TextStyle(  
        fontSize: 24,  
        fontWeight: FontWeight.bold,  
        color: Colors.white,  
      ), // TextStyle  
    ), // Text  
  
    const Text(  
      'BSIT 3 - 1\n',  
      style: TextStyle(  
        fontSize: 24,  
        color: Colors.white,  
      ), // TextStyle  
    ), // Text  
  ],  
), // Row
```



🚩 Flag 5 – Give It Some Space

Objective: Improve the spacing between your widgets.

Research:

`SizeBox()`

Add appropriate spacing between your:

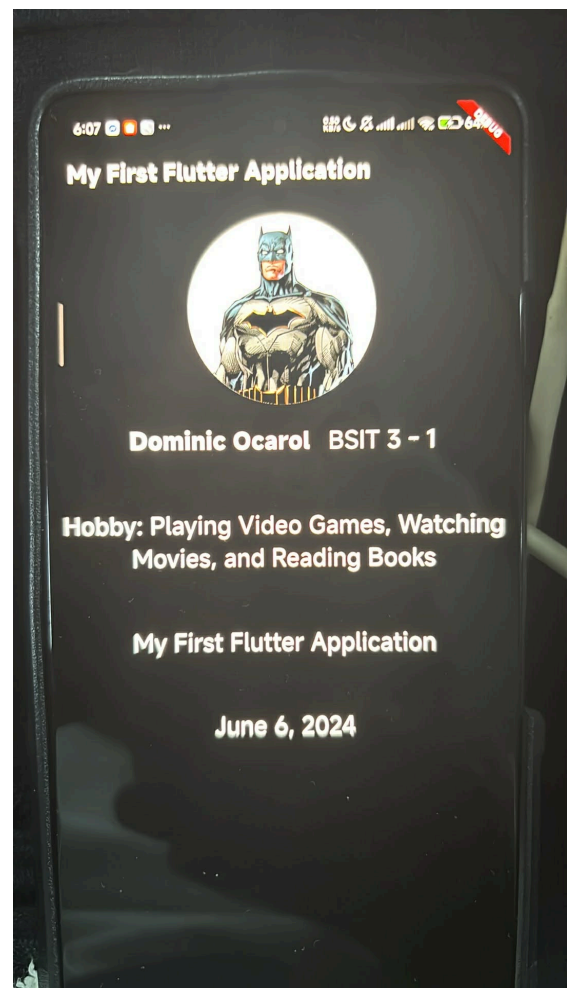
- Image
- Name
- Course
- Other information
- Row

Challenge: Make the card easier to read **without changing its colors or fonts yet.**

Screenshot:

- `SizeBox` usage
- Updated application

```
1 const SizeBox(height: 20),
2
3 Row(
4   mainAxisAlignment: MainAxisAlignment.center,
5   children: [
6     const Text(
7       'Dominic Ocarol\n',
8       style: TextStyle(
9         fontSize: 24,
10        fontWeight: FontWeight.bold,
11        color: Colors.white,
12      ), // TextStyle
13    ), // Text
14    const SizeBox(width: 20),
15
16    const Text(
17      'BSIT 3 - 1\n',
18      style: TextStyle(
19        fontSize: 24,
20        color: Colors.white,
21      ), // TextStyle
22    ), // Text
23  ],
24 ), // Row
25
26 const SizeBox(height: 20),
27
28 // Hobby
29 const Text(
30   'Hobby: Playing Video Games, Watching Movies, and Re
31   style: TextStyle(
32     fontSize: 24,
33     color: Colors.white,
34   ), // TextStyle
35   textAlign: TextAlign.center,
36 ), // Text
37
38 const SizeBox(height: 20),
39
```



🚩 Flag 6 – Make It Look Better

Objective: Apply styling to your individual widgets.

Now that your information is properly separated and positioned, improve the visual hierarchy.

Research how to modify:

`TextStyle()`

Apply different styling to at least **three** elements.

For example:

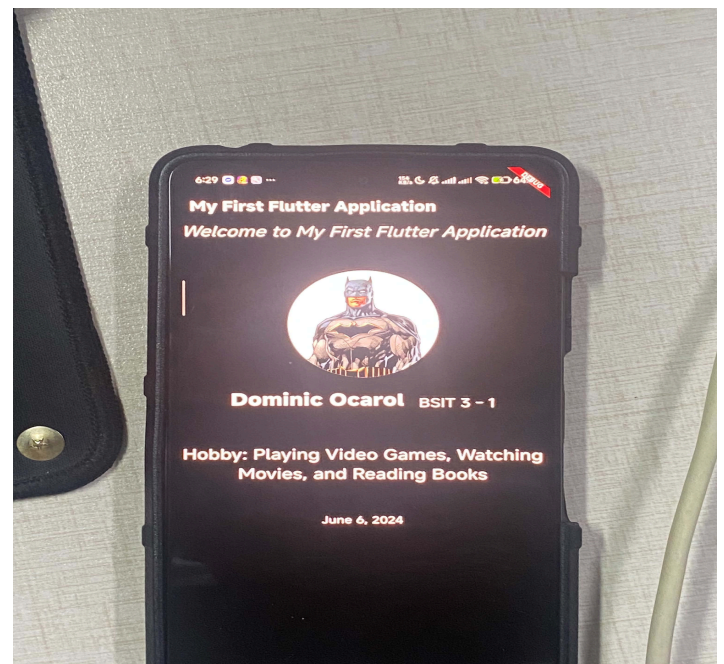
- Name → larger and bold
- Course → medium size
- Date → smaller
- Application title → italic

Screenshot:

- Your `TextStyle` changes

```
body: Column(  
  mainAxisAlignment: MainAxisAlignment.start,  
  children: [  
  
    const Text(  
      'Welcome to My First Flutter Application\n',  
      style: TextStyle(  
        fontSize: 24,  
        color: Colors.white,  
        fontStyle: FontStyle.italic,  
      ), // TextStyle  
      textAlign: TextAlign.center,  
    ), // Text  
  
    const SizedBox(height: 20),  
  
    CircleAvatar(  
      radius: 100,  
      backgroundColor: Colors.white,  
      backgroundImage: const AssetImage(  
        'assets/images/batman.jpg',  
      ), // AssetImage  
    ), // CircleAvatar  
  
    const SizedBox(height: 20),  
  
    Row(  
      mainAxisAlignment: MainAxisAlignment.center,  
      children: [  
        const Text(  
          'Dominic Ocarol\n',  
          style: TextStyle(  
            fontSize: 30,  
            fontWeight: FontWeight.bold,  
            color: Colors.white,  
          ), // TextStyle  
        ), // Text  
        const SizedBox(width: 20),  
        const Text(  
          'BSIT 3 - 1 application',  
          style: TextStyle(  
            fontSize: 20,  
            color: Colors.white,  
          ), // TextStyle  
        ),  
      ],  
    ),  
  ],  
)
```

```
// Date  
const Text(  
  'June 6, 2024',  
  style: TextStyle(fontSize: 15, color: Colors.white),  
  textAlign: TextAlign.center,  
) // Text
```



🚩 Flag 7 – Improve the Layout

Objective: Control the alignment of your widgets.

Research the properties of `Column` and `Row`.

Experiment with:

`mainAxisAlignment`

`crossAxisAlignment`

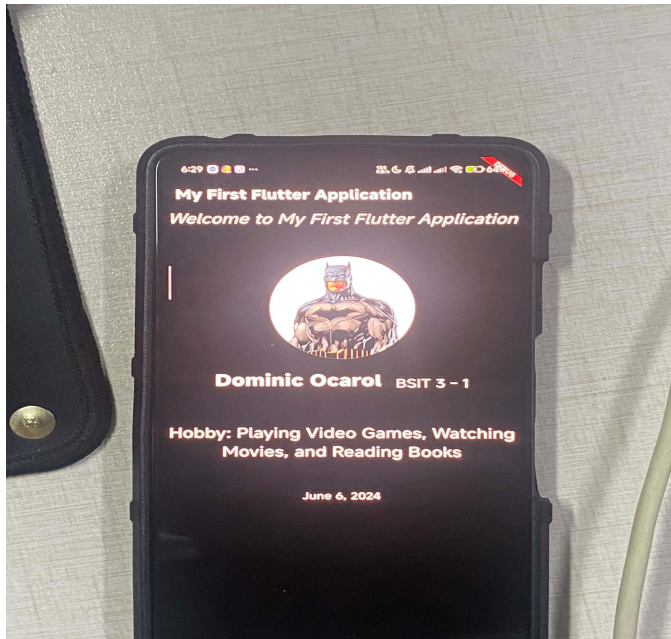
Change the alignment of your content so the card looks intentional rather than simply stacked.

Screenshot:

- The properties you changed

```
9
10
11 body: Column(
12   mainAxisAlignment: MainAxisAlignment.start,
13   children: [
14     const Text(
15       'Welcome to My First Flutter Application\n',
16       style: TextStyle(
17         fontSize: 24,
18         color: Colors.white,
19         fontStyle: FontStyle.italic,
20       ), // TextStyle
21       textAlign: TextAlign.center,
22     ), // Text
23
24     const SizedBox(height: 20),
25
26     CircleAvatar(
27       radius: 100,
28       backgroundColor: Colors.white,
29       backgroundImage: const AssetImage('assets/image
30     ), // CircleAvatar
31
32     const SizedBox(height: 20),
33
34     Row(
35       mainAxisAlignment: MainAxisAlignment.center,
36       children: [
```


- Updated application



🚩 Flag 8 – Create a Second Card

Objective: Apply what you learned to create another section.

Create a second **Card** containing something different from your profile.

For example:

My Favorites

Favorite Game

Favorite Food

Favorite Movie

The second card should also use:

- Individual **Text** widgets
- A **Column**
- At least one **Row**

- Spacing

Screenshot:

- Code for the second card

```
// Favorite Movie and Food
Card(
  child: Padding(
    padding: const EdgeInsets.all(8.0),
    child:
      const Text(
        'My favorite movie is: The Dark Knight\n',
        style: TextStyle(
          color: Colors.black,
          fontSize: 12,
          fontWeight: FontWeight.bold
        ), // TextStyle
      ), // Text
    ), // Padding
  ), // Card

// Favorite Food
Card(
  child: Padding(
    padding: const EdgeInsets.all(8.0),
    child:
      const Text(
        'My favorite Food is: Adobo\n',
        style: TextStyle(
          fontSize: 12,
          fontWeight: FontWeight.bold,
          color: Colors.black), // TextStyle
      ), // Text
    ), // Padding
  ), // Card
```

```
// Favorite Game
Row(
  mainAxisAlignment: MainAxisAlignment.center,
  children: [
    Column(
      children: const [
        Text(
          'My favorite game is: Dota2 , Valorant',
          style: TextStyle(
            fontSize: 20,
            fontWeight: FontWeight.bold,
            color: Colors.white
          ), // TextStyle
        ), // Text
      ],
    ), // Column
  ],
), // Row
```


🚩 Flag 9 – Stack the Cards

Objective: Arrange your two cards vertically.

You now have:

Profile Card



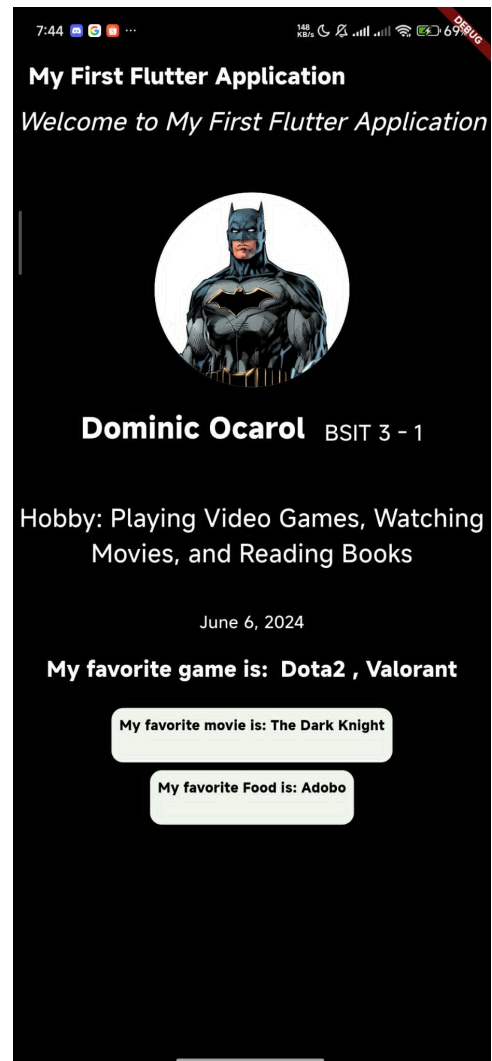
Favorites Card

Research how to place **multiple widgets inside the Flutter body**.

Use a layout widget to arrange your two cards vertically.

Screenshot:

- The widget containing both Cards
- Both cards visible in the application
-



🚩 Flag 10 – Widget Discovery

Objective: Discover a Flutter widget independently.

Research the Flutter widget catalog and find **one widget you haven't used yet**.

Examples:

- `Icon`
- `Divider`
- `CircleAvatar`
- `Chip`
- `ElevatedButton`

Add it somewhere appropriate in your application.

Screenshot:

- Your newly discovered widget in the code
- The widget visible in the application

```
CircleAvatar(  
  radius: 100,  
  backgroundColor: Colors.white,  
  backgroundImage: const AssetImage('assets/images/batman.jpg'),  
), // CircleAvatar
```

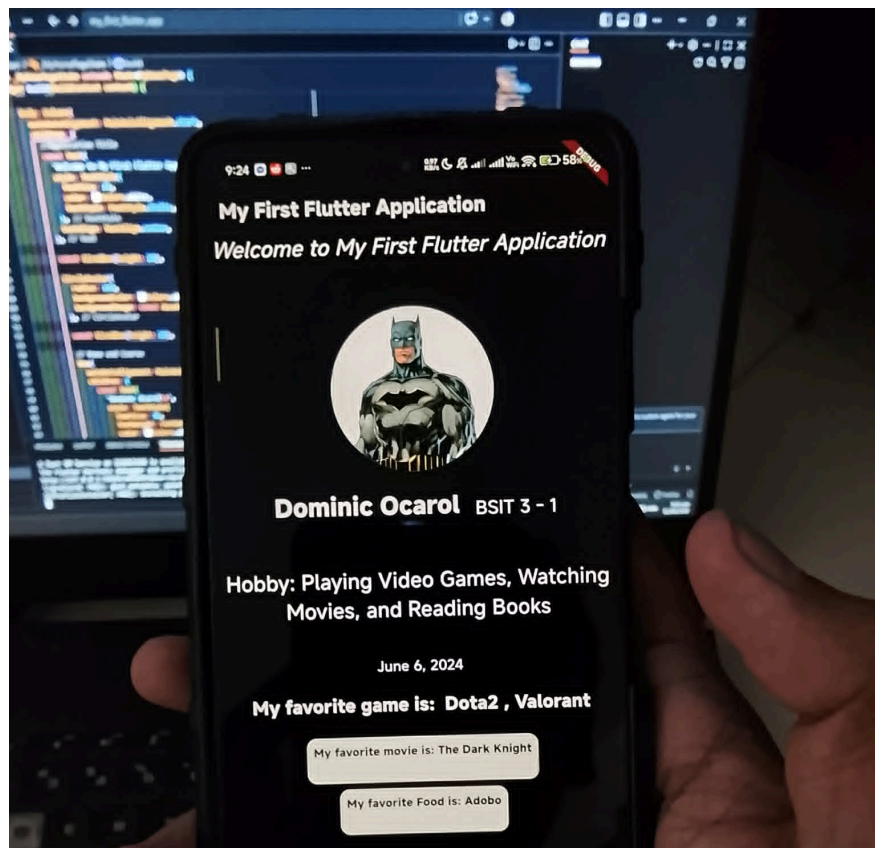
🚩 Flag 11 – Real Phone Test 📱

Objective: Run your completed application on a **physical** Android phone.

Research how to run Flutter on your phone.

Screenshot:

- Your completed application running on the physical phone (screenshot from your actual phone)



Flag 12 – Save Your Progress

Objective: Commit your completed Widget branch.

Follow proper standards of commit messages

Screenshot:

- Successful commit
- `feature/widgets` branch

```
dominic@NicoMac MINGW64 ~/source/repos/IT/IT314AB_Ocarol (feature/widgets)
$ git push origin feature/widgets
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 16 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 1.42 KiB | 1.42 MiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'feature/widgets' on GitHub by visiting:
remote:   https://github.com/domsprog/IT314AB_Ocarol/pull/new/feature/widgets
remote:
To https://github.com/domsprog/IT314AB_Ocarol.git
 * [new branch]      feature/widgets -> feature/widgets
```